



VER MONITORING REPORT

Guangdong Lankou 26MW Hydro Power Project

CDM Registration Reference no.2458

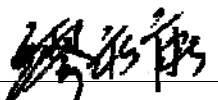
CDM Registration date: 07/07/2009 00:00:00

VER Monitoring Period:

01/11/2007 - 06/07/2009

Emission reduction in the monitoring period: 82,009 tCO₂e

Responsible person: Mr. Miao Yongkang

Signature: 



VER MONITORING REPORT FORM

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1 Project introduction

Guangdong Lankou 26MW Hydro Power Project (hereafter the project) has been registered by the UNFCCC on 7th, Jul, 2009 with the reference no. 2458. The project is located at Bainitang village, Lankou Town, Dongyuan County, Heyuan City where the geographical coordinate are east longitude 115°02'44" and north latitude 23°55'56". The operating time of the project is 1st, Nov, 2007.

Further information on the project could be seen available on the UNFCCC website:

<http://cdm.unfccc.int/Projects/DB/JCI1238492309.45/view>

2 Conformity with VCS requirements

The project claims emission reduction prior to CDM registration date under VCS 2007.1 standard including VCS 2007.1, further guidance for projects that are registered in two GHG programs, and ACM0002 Ver 7.

2.1. Project starting date

According to VCS 2007.1 clause 5.2.1, the Project Start Date was 1st, Nov, 2007.

According to VCS 2007.1 clause 5.2.1, the project validation had been completed within two years of the Project Start Date.

2.2. Project scope

The project applies the CDM methodology ACM0002 Ver. 7 which is approved in the VCS program.

2.3. Demonstration to confirm that the project was not implemented to create GHG emissions primarily for the purpose of its subsequent removal or destruction (according to VCS Project Description Template 1.12)

The project generates renewable energy without any project emission. Then the project hasn't been implemented to create GHG emissions.

2.4. Demonstration that the project has not created another form of environmental credit (for example renewable energy certificates, according to VCS Project Description Template 1.13)

The project proponent confirmed that the project has not created emission reductions other than VCU's (before registered as a CDM project) and CERs (after registered as a CDM project).



2.5. Project rejected under other GHG programs (if applicable, according to VCS Project Description Template 1.14)

The project just applied CDM and has been registered, it is not rejected under other GHG programs.

2.6. Proof of Title (according to VCS Project Description Template 8.1)

Project approval, Chinese LoA and business registry all show that Lankou project is a project owned by the Project Participant. The project therefore complies with paragraph 8.1 of the Project Description Template.

2.7. Projects that reduce GHG emissions from activities that participate in an emissions trading program (if applicable, according to VCS Project Description Template 8.2):

The Emission reduction will take place in CER trading program after the project is registered in and the Emission reduction before the CDM registration apply VCUs.

3 Key Monitoring Activities

3.1. Parameters Monitored



Table 1. Detailed information of parameters monitored

ID number	Data type	Data variable	Value	Data unit	Recording frequency	Proportion of data monitored	How will data be archived? (electronic/ paper)	For how long is archived data kept?	Comment
1. $EG_{output,y}$	Electricity quantity	Electricity delivered to the grid	94,328.5	MWh	Hourly measurement, monthly recorded	100%	Electronic and paper backup	Two years	Electricity supplied by the project to the grid. Double check by receipt of sales.
2. $EG_{input,y}$	Electricity quantity	Electricity imported from the grid	194.7	MWh	Hourly measurement, monthly recorded	100%	Electronic and paper backup	Two years	Electricity supplied by the grid to the project. Double check by balance ticket of purchases.
3. Cap_PJ	Capacity	Installed Capacity of the project	26,000,000	W	Measured yearly	100%	----	During the Crediting Period	Checked by the nameplate of the generators.
4. A_{PJ}	m ²	surface area at full reservoir level	141,441	m ²	Measured yearly	100%	Paper	During the Crediting Period	----
5. TEG_y	Electricity quantity	Total electricity produced by the project activity	101,963	MWh	Hourly measurement, monthly recorded	100%	Electronic and paper backup	Two years	----

3.2. Monitoring organization

Prior to the start of the crediting period, the organization of the monitoring team had been established. Clear roles and responsibilities have been assigned to all staffs involved in the project and a single CDM Manager has been nominated. The CDM Manager is overall responsible for the monitoring system on this project.

The structure of the monitoring organization is shown in the figure 1.

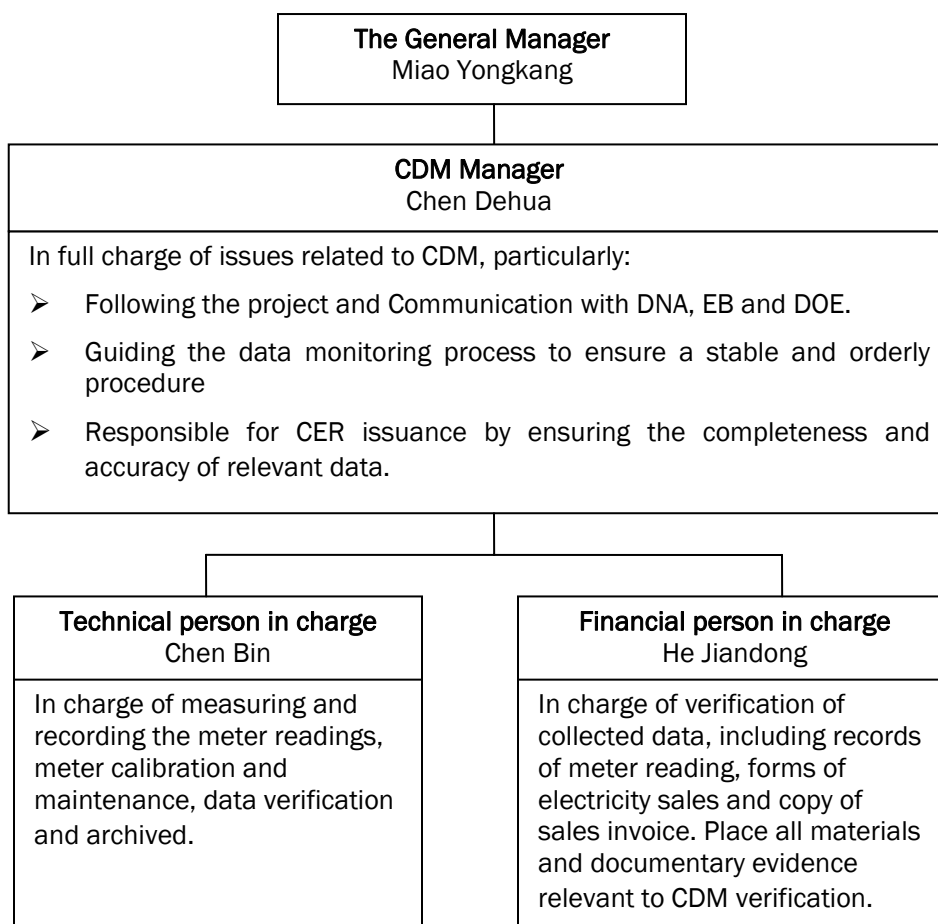


Figure 1 Structure of the Monitoring Organization

3.3. Staff training

Prior to commission of the Project, seminars were held to train the monitoring staffs to understand the requirements and processes of both operation and VCS monitoring. Relevant training materials and training records was documented for verification.

3.4. QA/QC

The quality control and quality assurance measures of the project are listed in the Table 2:

Table 2: Quality control and quality assurance measures

Parameter	Uncertainty level of data (High/Medium/Low)	QA/QC procedures
$EG_{output,y}$	Low	The data were monitored by the main electricity meter (M1) which is the revenue meter installed at the Lankou 110kV Transformer Station. A backup meter (M2) located at the exit of the booster station of the proposed project provided a useful cross check allowing for transmission losses as well as the electricity figures on the receipts. Electricity meters have met the relevant local standards at the time of installation and been calibrated according to the relevant regulations.
$EG_{input,y}$	Low	The data were monitored by the main electricity meter (M1) which is the revenue meter installed at the Lankou 110kV Transformer Station. A backup meter (M2) located at the exit of the booster station of the proposed project provided a useful cross check allowing for transmission losses as well as the electricity figures on the receipts. Electricity meters have met the relevant local standards at the time of installation and been calibrated according to the relevant regulations.
3.Cap _{PJ}	Low	Monitored yearly through the nameplate of the generators.
4.A _{PJ}	Low	Monitored yearly using the design schematics and area maps.
5.TEG _y	Low	The data were monitored by the electricity meters (M3, M4 and M5) which are installed at the exits of the generators of the project to measure the total electricity produced by the project activity TEG _y . Electricity meters have met the relevant local standards at the time of installation and been calibrated according to the relevant regulations.

3.5. Calibration information

The main meter (M1) located at the Lankou 110kV Substation and the backup meter (M2) located at the input of transmission line both have been calibrated before installation and after used according to the relevant regulations. Meters were duly calibrated by accredited agencies using applicable national standards and all reports are kept for verification.

4 Calculation of GHG Emission Reductions

4.1. Monitored Baseline emission

The baseline emission BE_y (tCO₂e) during the monitoring period results from:

$$BE_y = EG_y * EF_y$$



Where,

EG_y -Net electricity supplied to the grid during the monitoring period (MWh);

$$EG_y = EG_{output,y} - EG_{input,y}$$

According to Table 1, during the crediting period:

$$EG_{output}=94,328.5 \text{ MWh}, \quad EG_{input} = 194.7 \text{ MWh}$$

$$\begin{aligned} \text{Then } EG_y &= 94,328.5 \text{ MWh} - 194.7 \text{ MWh} \\ &= 94133.8 \text{ MWh} \end{aligned}$$

EF_y -Emission factor of the grid (tCO₂e /MWh)

According to the registered PDD, $EF_y = 0.8712 \text{ tCO}_2\text{e/MWh}$

$$BE_y = EG_y * EF_y$$

$$\begin{aligned} &= 94133.8 \text{ MWh} * 0.8712 \text{ tCO}_2\text{e/MWh} \\ &= 82,009 \text{ tCO}_2\text{e} \end{aligned}$$

4.2. Monitored Project Emission

The power density monitored is $183.82 \text{ W/m}^2 > 10 \text{ W/m}^2$, according to ACM0002 Ver.7, $PE_y = 0 \text{ tCO}_2\text{e}$.

4.3. Project leakage:

As per ACM0002 Ver 7, the project leakage is zero, i.e. $LE_y = 0 \text{ tCO}_2\text{e}$.

4.4. Monitored emission reductions

Parameter name	Item	symbol	Unit	Value
Baseline emission	(1)	BE_y	tCO ₂ e	82,009
Project emission	(2)	PE_y	tCO ₂ e	0
Leakage	(3)	LE_y	tCO ₂ e	0
Emission reduction	(4)=(1)-(2)-(3)	ER_y	tCO ₂ e	82,009